

EasyScience: A Framework for Unified Neutron Scattering Data Analysis



ANDREW
SAZONOV



PIOTR
ROZYCZKO



CHRISTIAN
VEDEL



HENRIK
JACOBSEN



ALES
KUTSEPAU

Framework



EasyScience (easyscience.org) is a framework initiated at the European Spallation Source (ess.eu) to help unify and simplify data analysis across neutron scattering techniques.

It aims to make the analysis step more accessible by providing user-friendly graphical interfaces for new users, while still supporting flexible script-based workflows in Jupyter notebooks for more advanced users.

Core Modules



EasyScience core library - building blocks for libraries and applications

backend



Graphical user interface components for desktop applications

frontend

Domain-Specific Projects



Diffraction data analysis desktop application & python library

model refinement



Bragg-Edge Imaging data analysis desktop application & python library

model refinement



Molecular structure generation desktop application

multiscale simulation workflows



Reflectometry data analysis desktop application & python library

model refinement



Quasielastic neutron scattering data analysis desktop application & python library

model refinement



Texture data reduction desktop application

data reduction



Main Features



Free and Open-Source

EasyScience applications and libraries are freely available for use. The source code is openly shared on github.com/easyscience



Cross-Platform

EasyScience-based applications run on all major operating systems: macOS, Windows, and Linux.



Improved User Experience

The graphical interface has undergone multiple improvements. Reviewed by UX experts from Design Psychology.



Reusable Components

A modular architecture enables reuse of generic components across techniques, improving consistency and maintainability.



Integrates External Calculators

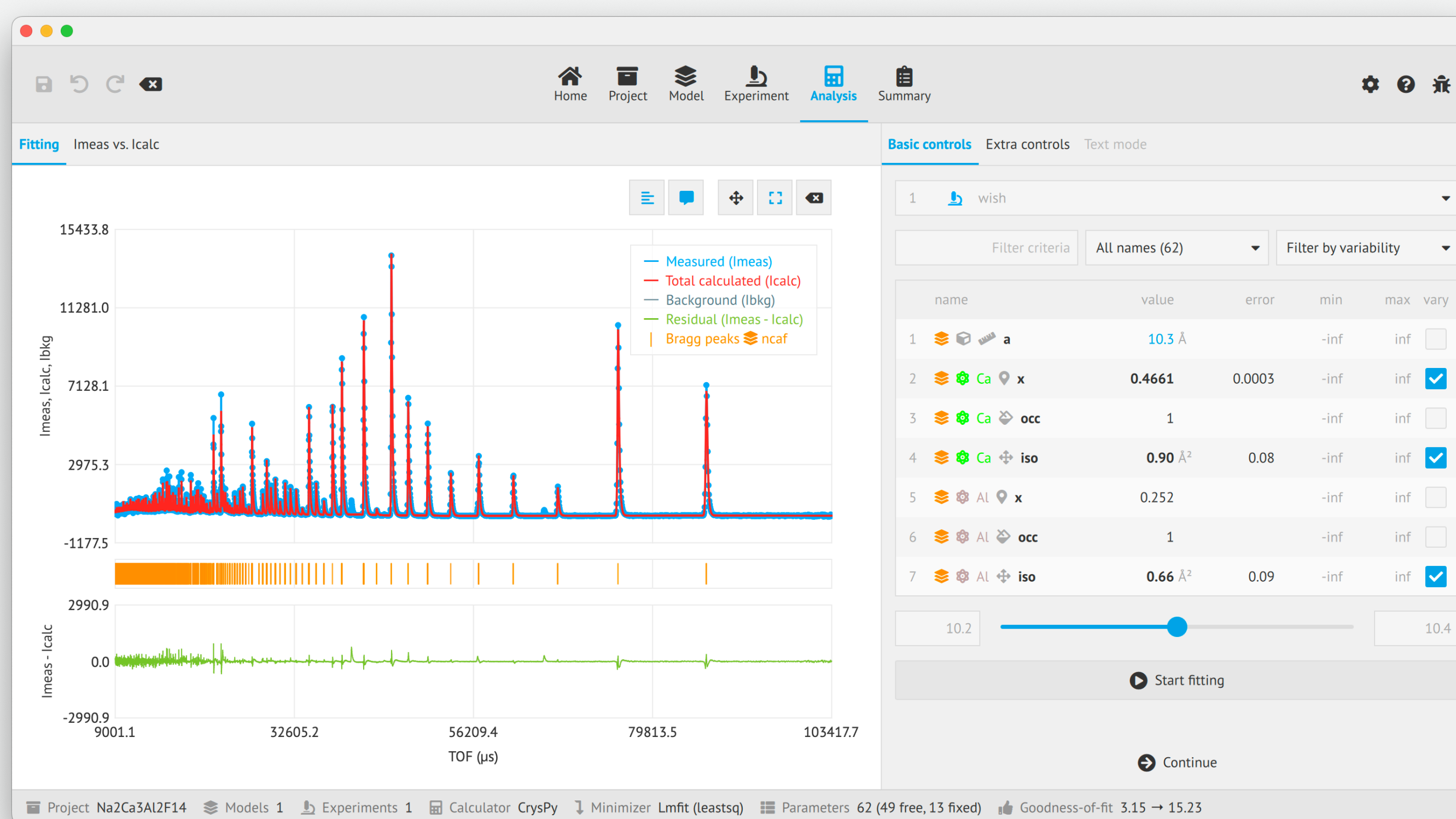
EasyScience simplifies the integration of domain-specific scientific libraries as external calculation engines.



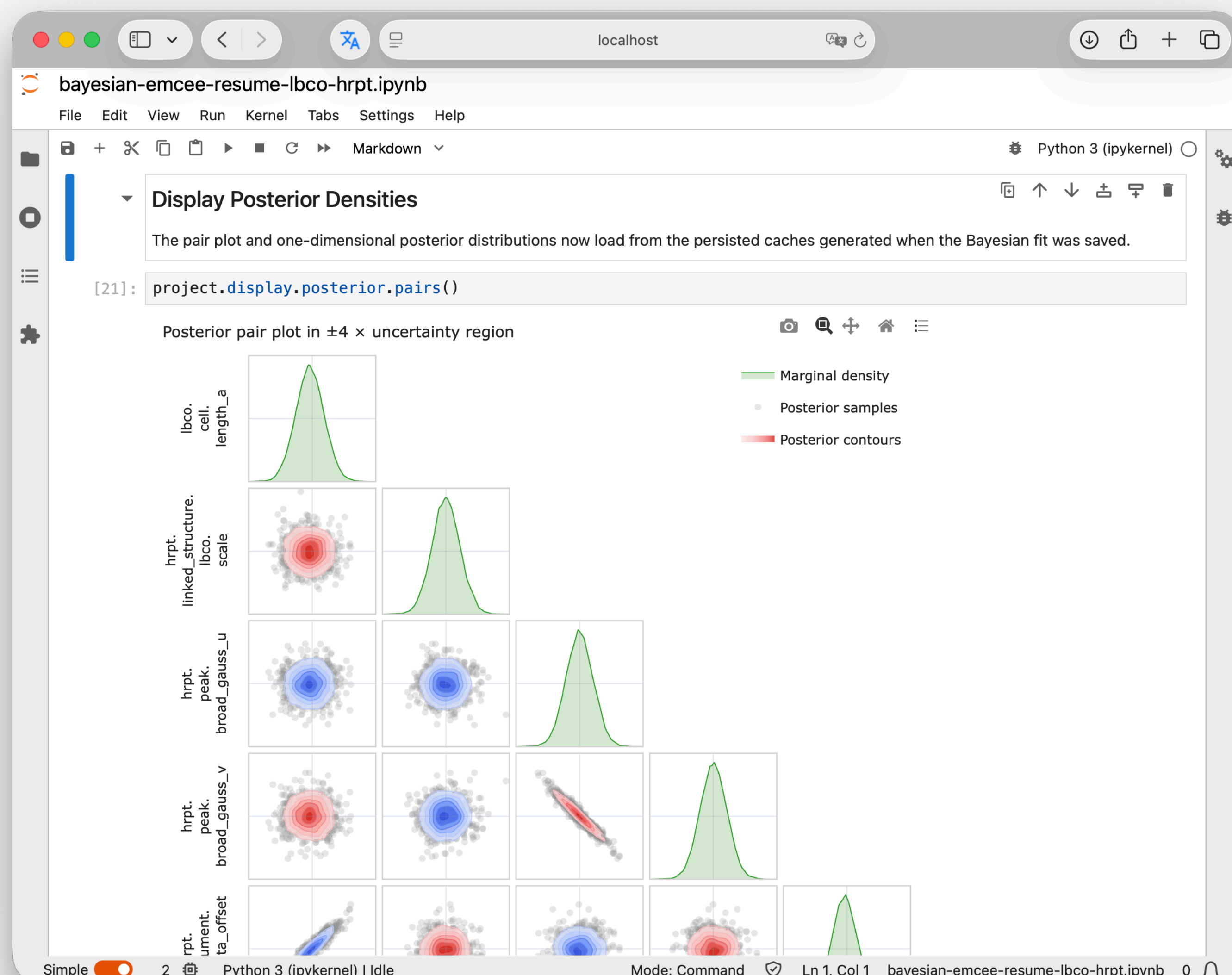
Adaptable to Different User Groups

EasyScience supports users with different data analysis preferences through both graphical and scripting interfaces.

Graphical Interface Example



Jupyter Notebook Example



Graphical App and Python Library

The EasyScience framework supports the development of both graphical desktop applications and Python-based libraries.



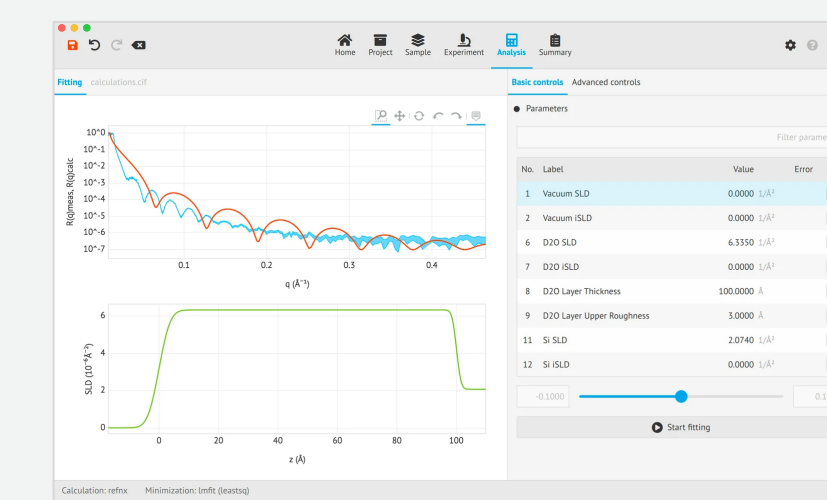
Consistent Look and Feel

Shared components ensure a uniform, user-friendly design across domain-specific applications, while allowing focus on scientific features.

EasyDiffraction



EasyReflectometry



Open for Contribution

We are open to collaboration!
Current contributors include
U Göttingen, JCNS @ MLZ, Chalmers U



Feedback

We would love to hear your feedback!
easyscience.org/#contact

